

in Azure. It provides: (a) Simple network deployment and operations; (b) Consortium management; (c) Smart contracts development with development tools.

It provides support for Ethereum Quorum ledger using the Istanbul Byzantine Fault Tolerance (IBFT) consensus mechanism. With these features, no administration is required, and users can concentrate on app development and business logic rather than allocating time and resources to managing virtual machines and infrastructure. It supports application development via open source tools and has an ample choice of platforms to deliver solutions.

Oracle Cloud

Oracle Blockchain Platform gives users a pre-assembled platform for building and running smart contracts and maintaining a tamper-proof distributed ledger.

This platform is a network consisting of validating nodes (peers) that update the ledger and respond to queries by executing smart contract code — the business logic that runs on the blockchain. External applications invoke transactions or run queries through client SDKs or REST API calls, which prompts selected peers to run the smart contracts. Multiple peers endorse (digitally sign) the results, which are then verified and sent to the ordering service. After a consensus is reached on the transaction order, transaction results are grouped into cryptographically secured, tamper-proof data blocks and sent to peer nodes to be validated and appended to the ledger. Service administrators can use the Oracle Blockchain Platform Web console to configure the blockchain and monitor its operation. With this platform, developers can complete some simple instance creation steps. Then Oracle takes care of service management, patching, backup and restoring, and other service life cycle tasks.

Oracle Blockchain Platform is available on Oracle Cloud Infrastructure (OCI), and can access other required services like Oracle Cloud Infrastructure Compute, Oracle Cloud Object Storage, and Oracle Identity Cloud Service.

Oracle Cloud Infrastructure Compute: The Oracle Blockchain Platform uses an Oracle Cloud Infrastructure Compute VM to deploy and run the Oracle Blockchain Platform instance and all other required applications such as Oracle Cloud Infrastructure Object Storage, Oracle Identity Cloud Service, and Oracle Cloud Infrastructure Load Balancing.

Oracle Cloud Infrastructure Object Storage: The Oracle Blockchain Platform uses Oracle Cloud Infrastructure Object Storage to store product-related binary files and logs.

Oracle Identity Cloud Service: Oracle Identity Cloud Service Foundation is automatically provided when you subscribe to the Oracle Blockchain Platform through Oracle Universal Credits. Some additional features are available with basic and standard editions.

Amazon Web Services (AWS)

AWS based blockchain has multi-flavour instances, like Hyperledger Fabric and Ethereum.

AWS provides purpose-built tools to support distinct needs, whether you need a centralised ledger database that maintains an immutable and cryptographically verifiable record of transactions, or a multi-party, fully managed blockchain network that helps eliminate intermediaries. Developing blockchain and ledger applications is simpler, faster, and more efficient with AWS. Amazon Managed Blockchain eliminates the heavy lifting involved in the setup of blockchain networks by reducing 60 per cent of the time taken in hosting Hyperledger Fabric frameworks. Managed blockchain also makes it easy to operate

networks as it supports AWS CLI, AWS CloudFormation, and Amazon Cloudwatch logs. Amazon QLDB is two to three times faster than traditional frameworks; it also provides SQL-like operators and a document data model for handling transactions.

AWS has over 70+ validated blockchain solutions from partners who provide support to all major blockchain protocols, including Hyperledger Sawtooth, Corda, DAML, Ethereum, Quorum, Blockstack, Blockapps Strato, RSK, Kadena ScalableBFT, and many more.

Amazon Managed Blockchain

Amazon Managed Blockchain is a fully managed service that makes it easy to create and manage scalable blockchain networks using the popular open source frameworks Hyperledger Fabric and Ethereum. This service makes it possible to build applications where multiple parties can execute transactions without the need for a trusted, central authority. Today, building a scalable blockchain network with existing technologies is complex to set up and hard to manage. To create a blockchain network, each network member needs to manually provision hardware, install software, create and manage certificates for access control, and configure networking components. Once the blockchain network is running, you need to continuously monitor the infrastructure and adapt to changes, such as an increase in transaction requests or new members joining or leaving the network.

Amazon Managed Blockchain is a fully managed service that allows you to set up and manage a scalable blockchain network with just a few clicks. It eliminates the overhead required to create the network, and automatically scales to meet the demands of thousands of applications running millions of transactions. Once your network is up and running, Managed Blockchain makes it easy to